

Byeonghwi Kim

✉ byeonghwikim@snu.ac.kr

🔗 btkim94.github.io

in [byeonghwi-kim-821909167](#)

🔗 [btkim94](#)

Research Interest

I am interested in building **multi-modal embodied agents** that integrate perception, reasoning and control to understand and interact with humans and environments for physically grounded services.

Work Experience

Krafton AI | Research Intern @ Physical AI Team

• Mentor: *Dr. Kangwook Lee*

Seoul, South Korea

Jun. 2025 - Dec. 2025

Allen Institute for AI (AI2) | Research Intern @ PRIOR Team (Remote)

• Mentor: *Dr. Roozbeh Mottaghi*

Seattle, WA, USA

Sept. 2021 - Mar. 2022

Education

Ph.D. Seoul National University | Electrical and Computer Engineering

• Advisor: *Jonghyun Choi*

Sept. 2024 - Present

Ph.D. Yonsei University | Artificial Intelligence

Course • Advisor: *Jonghyun Choi*

Sept. 2022 - Aug. 2024

M.S. Gwangju Institute of Science and Technology (GIST) | Artificial Intelligence

• *GIST President Fellowship*

• Advisor: *Jonghyun Choi*

Mar. 2020 - Aug. 2022

B.S. Kwangwoon University | Computer Engineering

• Graduated with First-Class Honors (GPA: 4.42/4.50, Major GPA: 4.48/4.50)

• Advisor: *Cheolsoo Park*

Mar. 2013 - Feb. 2020

Military Service (2014-2015)

Publications

Zero-Shot Grasp Affordance Grounding via Generation 

Byeonghwi Kim, Taewoong Kim, Jimin Nam, Jaehong Min, Jinyeon Kim, Taewook Kang, Jaehong Kim, Hyoeun Kim, Hyejeong Jeon, Jonghyun Choi

Embodied Agent and Dialog (EAD) Workshop @ ECCV 2026

Multi-Modal Grounded Planning and Efficient Replanning for Learning Embodied Agents with A Few Examples

Taewoong Kim, **Byeonghwi Kim**, Jonghyun Choi

AAAI 2025 (Oral)

Pre-emptive Action Revision by Environmental Feedback for Embodied Instruction Following Agents

Jinyeon Kim*, Cheolhong Min*, **Byeonghwi Kim**, Jonghyun Choi

CoRL 2024

ReALFRED: An Embodied Instruction Following Benchmark in Photo-Realistic Environments

Taewoong Kim*, Cheolhong Min*, **Byeonghwi Kim**, Jinyeon Kim, Wonje Jung, Jonghyun Choi

ECCV 2024

Online Continual Learning for Interactive Instruction Following Agents

Byeonghwi Kim*, Minhyuk Seo*, Jonghyun Choi

ICLR 2024

Context-Aware Planning and Environment-Aware Memory for Instruction Following Embodied Agents

Byeonghwi Kim, Jinyeon Kim, Yuyeong Kim, Cheolhong Min, Jonghyun Choi

ICCV 2023

Multi-Level Compositional Reasoning for Interactive Instruction Following

Suvaansh Bhambri*, **Byeonghwi Kim***, Jonghyun Choi
AAAI 2023 (Oral)

Factorizing Perception and Policy for Interactive Instruction Following

Kunal Pratap Singh*, Suvaansh Bhambri*, **Byeonghwi Kim***, Roozbeh Mottaghi, Jonghyun Choi
ICCV 2021

Agent with the Big Picture: Perceiving Surroundings for Interactive Instruction Following

Byeonghwi Kim, Suvaansh Bhambri, Kunal Pratap Singh, Roozbeh Mottaghi, Jonghyun Choi
Embodied AI Workshop @ CVPR 2021

Automated White Blood Cell Counting in Nailfold Capillary Using Deep Learning Segmentation and Video Stabilization

Byeonghwi Kim*, Yuli Sun Hariyani*, Young-Ho Cho, Cheolsoo Park
Sensors 2020

Honors and Awards

Gold Prize: Outstanding Paper Awards (IPIU 2025, South Korea)	Feb. 2025
• <i>Zero-Shot Grasp Affordance Grounding via Generation</i>	
Silver Prize: Graduate School Innovation Outstanding Paper Awards (Yonsei University)	Jul. 2024
• <i>Context-Aware Planning and Environment-Aware Memory for Instruction Following Embodied Agents</i>	
Silver Prize: Outstanding Paper Awards (IPIU 2024, South Korea)	Feb. 2024
• <i>Multi-Modal Grounded Planning and Efficient Replanning for Learning Embodied Agents with A Few Examples</i>	
Bronze Prize: Outstanding Paper Awards (IPIU 2024, South Korea)	Feb. 2024
• <i>Online Continual Learning for Interactive Instruction Following Agents</i>	
Bronze Prize: Graduate School Innovation Outstanding Paper Awards (Yonsei University)	Jul. 2023
• <i>Multi-Level Compositional Reasoning for Interactive Instruction Following</i>	
1st Place: Generalist Language Grounding Agents Challenge (CVPRW'23)	Jun. 2023
• <i>ECLAIR: Event-Cognizant Language Interaction Robots</i>	
2nd Place: ALFRED Challenge (CVPRW'21)	Jun. 2021
• <i>Agent with the Big Picture: Perceiving Surroundings for Interactive Instruction Following</i>	
2nd Place: ALFRED Challenge (ECCVW'20)	Aug. 2020
• <i>Improving Mask Prediction for Long Horizon Instruction Following</i>	

Scholarships

YoulChon AI Star Scholarship (Nongshim Youlchon Foundation & SNU AI Institute)	2025
SNU EE-Alumni Scholarship for Graduate Study (SNU EE-Alumni Association)	2024-2025
GIST President Fellowship (Gwangju Institute of Science and Technology (GIST))	2020-2022
Academic Excellence Scholarship (Kwangwoon University)	2013-2020

Academic Service

Reviewer / Program Committee Member

- Computer Vision
 - IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 2024-2026
 - IEEE Transactions on Circuits and Systems for Video Technology (**TCSVT**), 2025
 - IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2023-2026
 - International Conference on Computer Vision (**ICCV**), 2025

- European Conference on Computer Vision (**ECCV**), 2024-2026
- ACM International Conference on Multimedia (**MM**), 2025-2026
- ACM International Conference on Multimedia (**MM**) - Datasets Track, 2025
- ACM International Conference on Multimedia (**MM**) - Demo & Video Track, 2026
- IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**), 2023-2027
- The British Machine Vision Conference (**BMVC**), 2025-2026
- ACM International Conference on Multimedia Asia (**MMAsia**), 2025-2026
- Machine Learning
 - Transactions on Machine Learning Research (**TMLR**), 2026
 - International Conference on Learning Representations (**ICLR**), 2024-2027
 - International Conference on Machine Learning (**ICML**), 2024-2026
 - * Recognized as a Silver Reviewer for ICML 2026
 - International Conference on Machine Learning (**ICML**) - Position Paper Track, 2025
 - Conference on Neural Information Processing Systems (**NeurIPS**), 2023-2024, 2026
 - Conference on Neural Information Processing Systems (**NeurIPS**) - Datasets & Benchmarks Track, 2025
 - Conference on Neural Information Processing Systems (**NeurIPS**) - Position Paper Track, 2025
 - International Conference on Artificial Intelligence and Statistics (**AISTATS**), 2025-2027
 - AAAI Conference on Artificial Intelligence (**AAAI**), 2024-2027
- Robotics
 - Conference on Robot Learning (**CoRL**), 2025-2026
 - IEEE Robotics and Automation Letters (**RA-L**), 2025
 - IEEE International Conference on Robotics and Automation (**ICRA**), 2022, 2025
 - IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2025
 - IEEE International Symposium on Robot and Human Interactive Communication (**RO-MAN**), 2024-2025

Presentation

- | | |
|--|-----------|
| Invited poster presentation AI SEOUL 2024 @ Seoul Metropolitan Government, South Korea | Feb. 2024 |
| <ul style="list-style-type: none"> • <i>Context-Aware Planning and Environment-Aware Memory for Instruction Following Embodied Agents</i> | |
| Invited talk Samsung AI Forum 2023 @ Samsung Research, South Korea | Nov. 2023 |
| <ul style="list-style-type: none"> • <i>Context-Aware Planning and Environment-Aware Memory for Instruction Following Embodied Agents</i> • <i>Story Visualization by Online Text Augmentation with Context Memory</i> | |
| Invited talk LG AI Research, South Korea | Jun. 2023 |
| <ul style="list-style-type: none"> • <i>Multi-Level Compositional Reasoning for Interactive Instruction Following</i> • <i>Agent with the Big Picture: Perceiving Surroundings for Interactive Instruction Following</i> • <i>Factorizing Perception and Policy for Interactive Instruction Following</i> | |
| Invited talk Electronic & Information Research Information Center, South Korea | May 2023 |
| <ul style="list-style-type: none"> • <i>Multi-Level Compositional Reasoning for Interactive Instruction Following</i> | |

Teaching Experience

- | | |
|---|--|
| Teaching Assistant @ Seoul National University | |
| <ul style="list-style-type: none"> • <i>Introduction to Algorithms</i> • <i>Deep Learning</i> • <i>Introduction to Random Variables and Random Processes</i> | Spring 2025, Fall 2026
Fall 2024
Spring 2024 |
| Teaching Assistant @ Yonsei University | |
| <ul style="list-style-type: none"> • <i>Deep Learning - Theory and Practice</i> | Fall 2022, Spring 2023, Fall 2023 |

Teaching Assistant @ Gwangju Institute of Science and Technology

- *Machine Learning and Deep Learning*

Spring 2021

Teaching Assistant @ Samsung Electronics

- *AI Expert Program (Continual Learning)*
- *AI Expert Program (Object Detection)*

2022-2026

2023

Teaching Assistant @ HD Hyundai

- *SNU AI Education (Projects with AI Assistance)*

2026

Teaching Assistant @ Woori Financial Group

- *AI/Data Expert Course (Transformer-Based OCR and Object Detection)*

2026

Teaching Assistant @ Korea Telecom (KT)

- *KT-SNU AI Training Course - Computer Vision*

2024

Teaching Assistant @ LG Electronics

- *AI Advanced Course for CTO (Object Detection, Continual Learning)*

2023

Teaching Assistant @ Samsung Advanced Technology Training Institute

- *AI Academy for Application - Vision (Object Detection, Continual Learning)*

2022-2023

Patents

Method of Continual Learning on Embodied Agent Model Capable of Receiving Language Instructions and Performing Tasks, Method of Testing the Embodied Agent Model, Learning Device and Testing Device Using the Same

Jonghyun Choi, **Byeonghwi Kim**, Minhyuk Seo

Korea Patent 10-2746049

Method for Performing Tasks According to CAPEAM Model Including Context-Aware Planning Module and Environment-Aware Memory Module and AI Agent Using the Same

Jonghyun Choi, **Byeonghwi Kim**, Jinyeon Kim, Cheolhong Min, Yuyeong Kim

Korea Patent 10-2675973

References

References will be provided upon request.